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DATA ANALYTICS AND VISUALIZATION(24DSB3101)

An Explainable AI-Driven Big Data Analytics Framework for Climate Prediction and Environmental Risk Assessment

Batch – 6

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Problem Statement:

- Climate change produces large and complex datasets involving temperature, rainfall, greenhouse gases, sea levels, extreme weather events, and other environmental indicators.
- Existing AI and machine-learning models can provide accurate predictions, but many of them operate as black-box models, making it difficult to understand why a particular climate or environmental risk prediction was generated.
- Therefore, there is a need for a scalable Big Data Analytics and Explainable AI (XAI) framework that can predict climate-related risks while providing understandable explanations of the factors influencing those predictions.

Project Objectives:

- **To identify** and visualize important climate and environmental trends, patterns, relationships, and risks through comprehensive data analysis.
- **To leverage** predictive analytics and explainable AI techniques to identify environmental risk levels and the factors influencing them.
- ***To enhance*** *the existing system, we propose a* ***Multi-Source Climate Intelligence Framework*** *that integrates weather, satellite, IoT, and historical disaster data using Big Data Analytics, Machine Learning, and Explainable AI for comprehensive environmental risk prediction.*
- **To provide** clear, interactive, and data-driven insights that can support environmental monitoring, planning, and informed decision-making.
- **To develop** an interactive data analytics dashboard for analyzing climate and environmental data from multiple sources.

Literature Survey:

- **Climate Prediction using AI/ML:** Previous studies have used machine learning algorithms such as Random Forest, SVM, and neural networks to predict climate variables like temperature, rainfall, and extreme weather events.
- **Big Data for Environmental Analysis:** Researchers have used large-scale climate and environmental datasets from satellites, sensors, and government sources to identify patterns and improve prediction accuracy.
- **Explainable AI (XAI):** Techniques such as SHAP, LIME, and feature importance have been applied to explain AI predictions, making climate-risk models more transparent and understandable.
- **Environmental Risk Prediction:** Recent research focuses on predicting risks such as floods, droughts, heatwaves, wildfires, and other climate-related disasters using AI-based models.
- **Research Gap:** Many existing models focus mainly on prediction accuracy, while limited work combines big-data analytics, climate prediction, environmental risk assessment, and explainability in a single framework. This creates an opportunity for an integrated XAI-based climate risk prediction system.

Research Gaps Identification:

- Many climate prediction models focus primarily on **prediction accuracy** rather than interpretability.
- Several AI-based climate models function as **black boxes**, making their decisions difficult for environmental researchers and policymakers to understand.
- Existing studies often focus on **one climate variable or one type of environmental risk** rather than combining multiple factors.
- Large-scale heterogeneous climate datasets create challenges in **data processing and scalability**.
- XAI techniques are not consistently integrated with **Big Data-based climate analytics**.
- Limited research combines **climate prediction + environmental risk assessment + explainability** in a single framework.
- Many studies evaluate models technically but provide limited explanation of **which environmental factors drive the predicted risk**.

Objective 1 — Integrate environmental datasets

- Combine information from:
  - Climate/environmental data,
  - Extreme-weather information,
  - Air-quality information.

Objective 2 — Preprocess the integrated data

Perform:

- cleaning,
- missing-value handling,
- feature selection,
- normalization/standardization,
- temporal feature engineering.

Objective 3 — Build ML models

Investigate models such as:

- Logistic Regression
- Random Forest
- HistGradientBoosting
- XGBoost

Objective 4 — Build the final hybrid model

Combine:

- Logistic Regression + XGBoost at the probability level.

Objective 5 — Improve probability reliability

Apply:

- Platt Scaling to calibrate the hybrid probability.

Objective 6 — Add historical environmental evidence

- Calculate dataset-derived disaster rates under selected environmental conditions.

Objective 7 — Detect unusual live observations

Use:

- Mahalanobis-distance-based OOD detection to determine whether a live observation resembles the training distribution.

Objective 8 — Explain predictions

Use:

- SHAP to identify feature contributions.

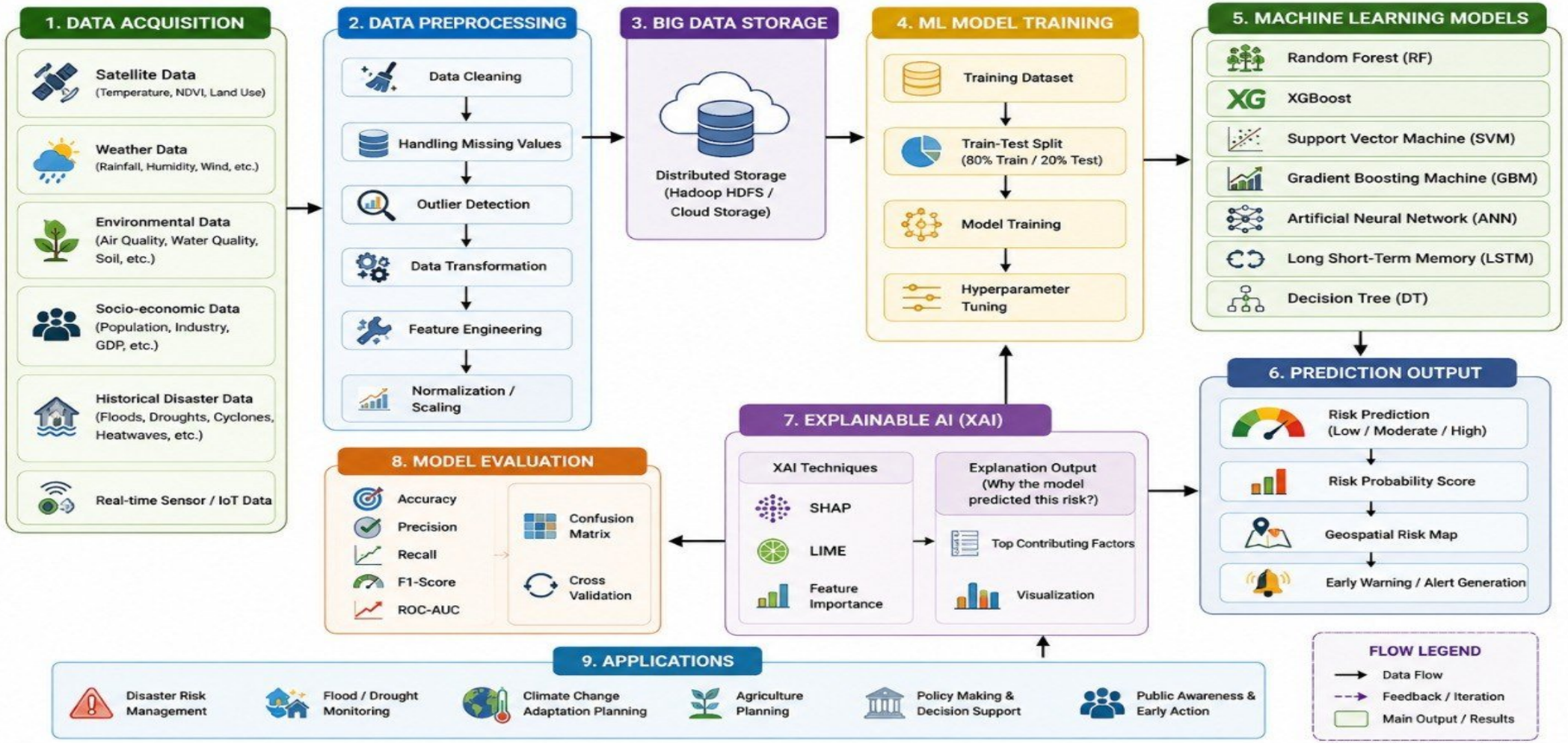
Objective 9 — Deploy the system

- Building an interactive Streamlit dashboard using live weather and air-quality APIs.

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Explainable AI for Climate Change Prediction and Environmental Risk Analysis Using Big Data Analytics



Three Datasets

Dataset 1 — Climate Dataset

dataset1_climate_cleaned.csv

- Size:
  - 2,425 rows
  - 13 columns
- This dataset forms the main historical modeling representation.

Dataset 2 — Extreme Weather Dataset

- File:
- dataset2_extreme_weather_cleaned.csv
- Size:
  - 49 records
  - 23 columns
- This dataset contributes extreme-weather/environmental information during the integration stage.

Dataset 3 — Air Quality Dataset

- File:
  dataset3_air_quality_cleaned.csv
- Size:
  - 842,160 hourly records
  - 28 columns
- Air-quality information is retained for:
  - environmental context,
  - dashboard display,
  - live environmental monitoring.

However:
Air-quality variables are NOT part of the final eight-feature ML model.

Data Integration

Climate information

+

Extreme-weather information

+

Air-quality information

↓

Cleaning

↓

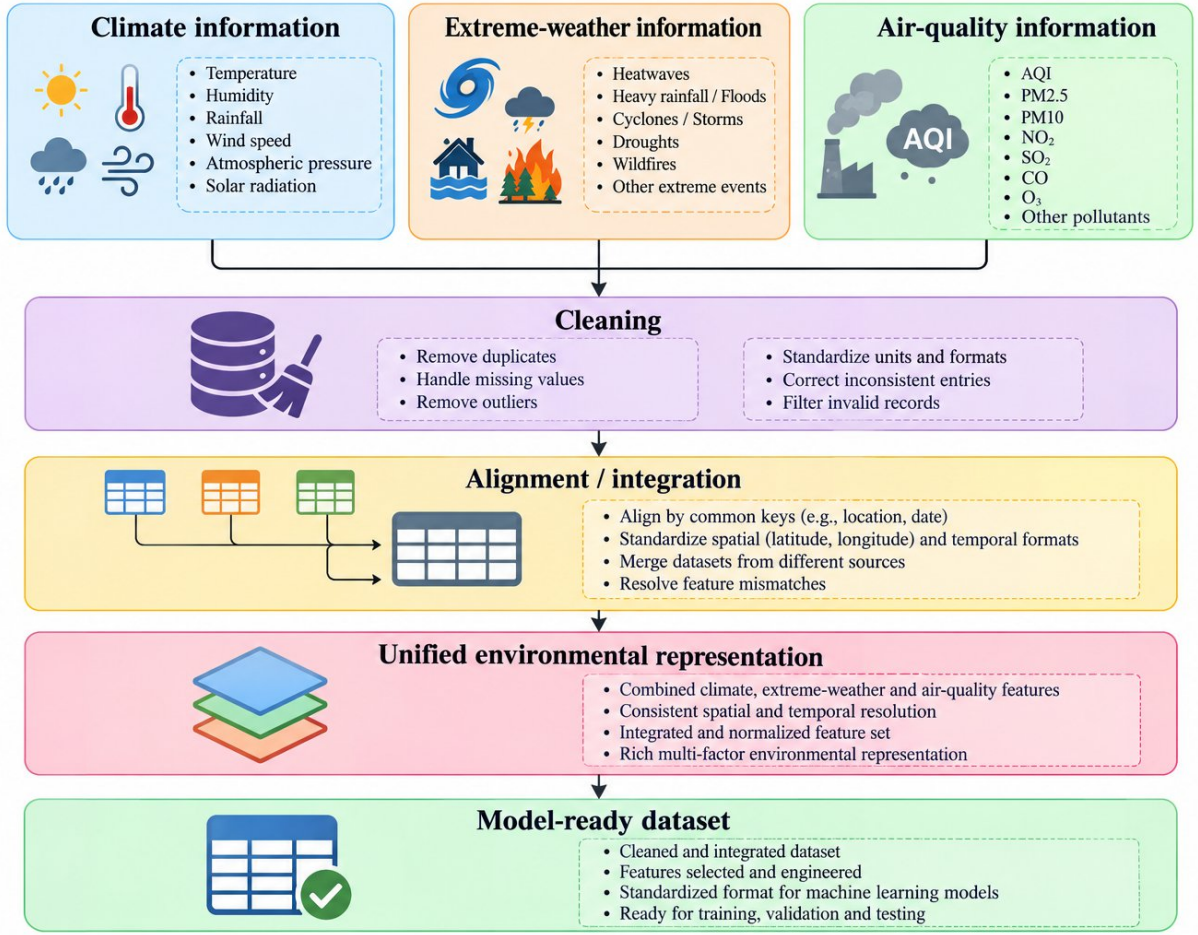
Alignment / integration

↓

Unified environmental representation

↓

Model-ready dataset



Data Preprocessing:

- Data preprocessing involved cleaning the integrated environmental datasets, handling missing numerical observations through median imputation, selecting the variables relevant to environmental risk assessment, and transforming the temporal information into cyclic features.
- The model inputs were standardized where required by the Logistic Regression pipeline to ensure comparable feature scales.
- Day-of-year was represented using sine and cosine transformations to preserve the cyclic nature of annual seasonality.
- Air-quality variables were retained for environmental context and dashboard visualization but were excluded from the final eight-feature machine-learning representation.
- The resulting model-ready dataset contained 2,425 observations and eight predictors.

Target Variable:

The target variable is: DISASTER

It is a binary classification problem:

0 → Normal / No Disaster

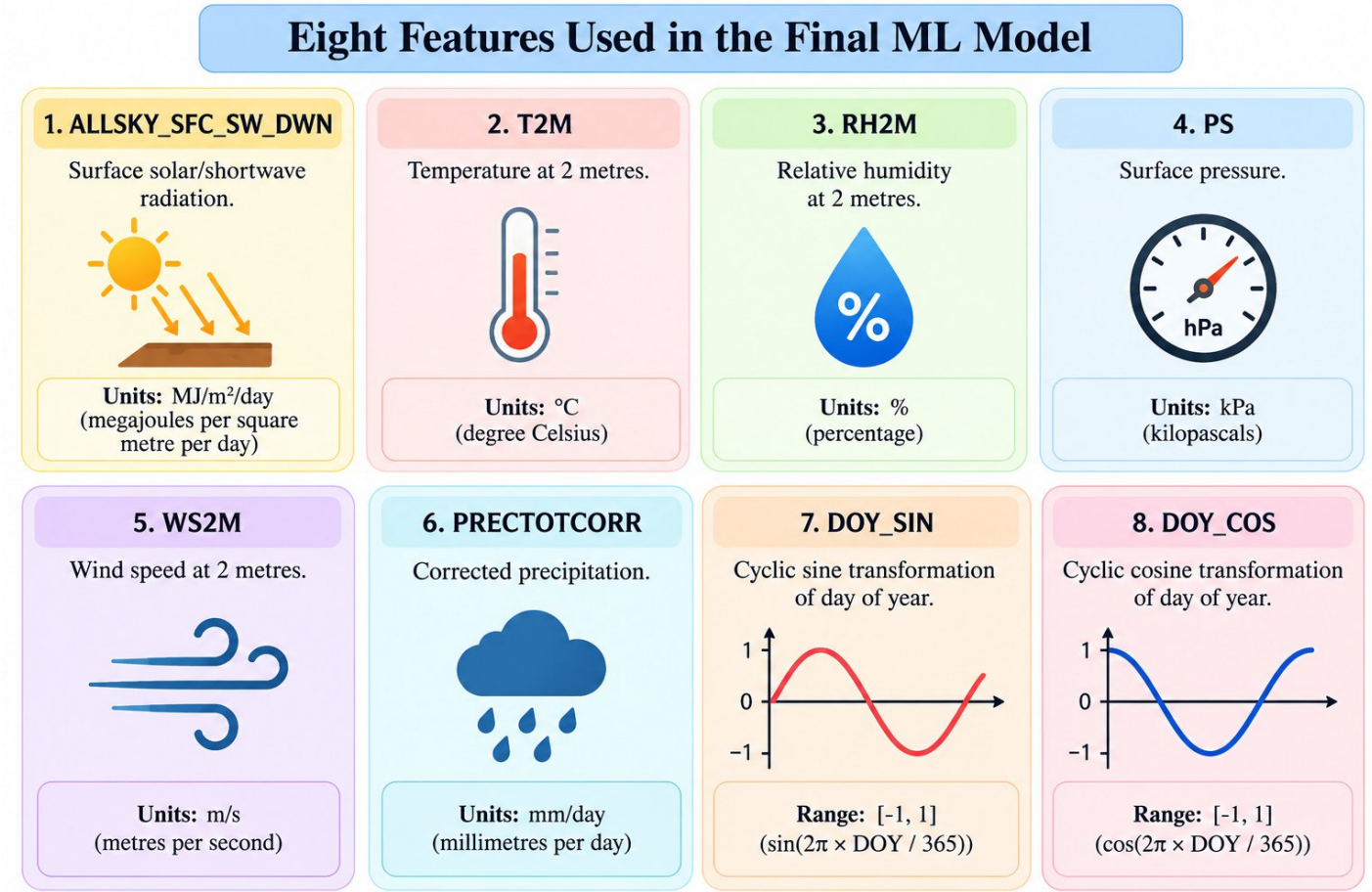
1 → Disaster

Final dataset:

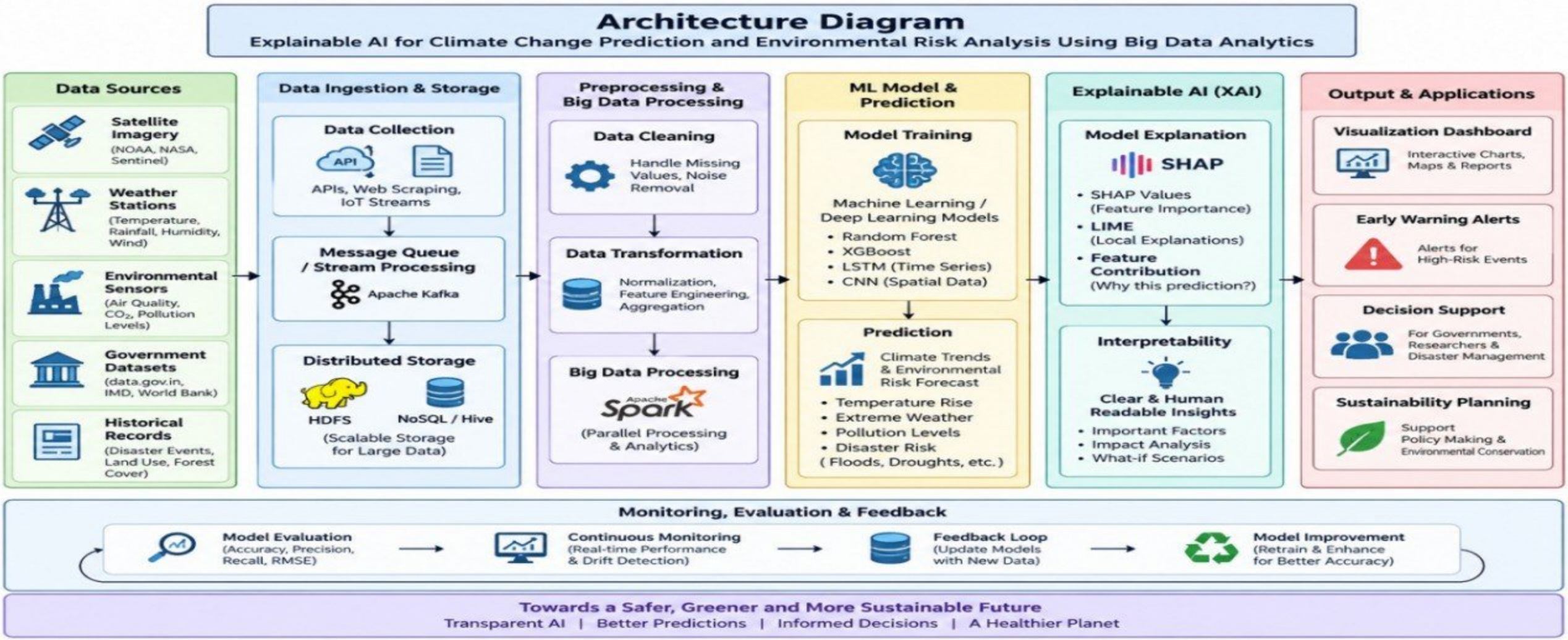
Class	Count	Percentage
Normal	2,172	89.57%
Disaster	253	10.43%
Total	2,425	100%

Final Eight ML Features

The final ML model uses exactly these eight features:
1. ALLSKY_SFC_SW_DWN
Surface solar/shortwave radiation.
2. T2M
Temperature at 2 metres.
3. RH2M
Relative humidity at 2 metres.
4. PS
Surface pressure.
5. WS2M
Wind speed at 2 metres.
6. PRECTOTCORR
Corrected precipitation.
7. DOY_SIN
Cyclic sine transformation of day of year.
8. DOY_COS
Cyclic cosine transformation of day of year.



Project Plan:



Proposed Work and Its Uniqueness:

- **Combines climate change prediction and environmental risk analysis** in one system.
- Uses **multiple environmental and climate factors** instead of focusing on a single hazard.
- Integrates **Big Data Analytics + Machine Learning + Explainable AI (XAI)**.
- Provides **explanations for predictions**, showing which factors contributed to the risk level.
- Aims to provide **risk classification and early-warning insights**, making the model more useful for real-world environmental decision-making.



Methodology:

1. Proposed Method / Approach

- Developed an Explainable AI-based Environmental Disaster Risk Prediction Framework.
- Combines historical environmental data, machine learning, risk scoring, and real-time weather data.
- The workflow includes data preprocessing → feature analysis → ML prediction → risk assessment → XAI → dashboard.

2. Dataset Collection & Data Preparation

- Collected and integrated three environmental datasets from different sources.
- Performed data cleaning, missing-value handling, feature selection, and normalization.
- **Combined relevant environmental parameters such as temperature, humidity, rainfall, wind speed, pressure.**
- Created additional seasonal/time-based features such as DOY_SIN and DOY_COS.
- Prepared the final dataset for disaster-risk classification.

Methodology:

3. ML Models & Model Combination

- Trained and evaluated multiple ML classification models on the prepared dataset.
- ML Models Used
  - Logistic Regression
  - Random Forest
  - Support Vector Machine (SVM)
  - Comparative evaluation of models
- ***Final Proposed Model: Hybridized Logistic Regression and Random Forest***
- Compared their performance using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix.
- Explored combinations/ensemble approaches using 2–3 models to improve prediction reliability.
- Analyzed the influence of environmental features on disaster predictions.

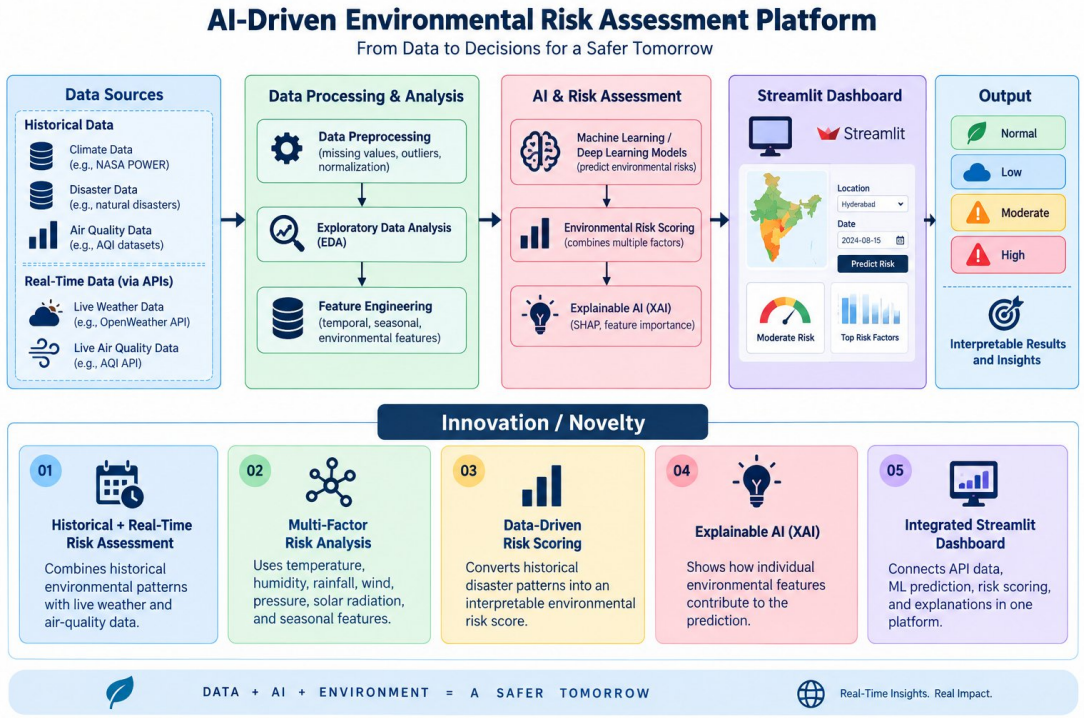
4. Final Proposed Model

- Selected the final Logistic Regression-based model as the proposed prediction model.
- Applied class balancing, median imputation, and feature standardization.
- ***Used a tuned 0.71 decision threshold for disaster classification.***
- Integrated the model with a historical environmental risk score + real-time weather API + XAI.
- Implemented the complete system through a Streamlit dashboard.

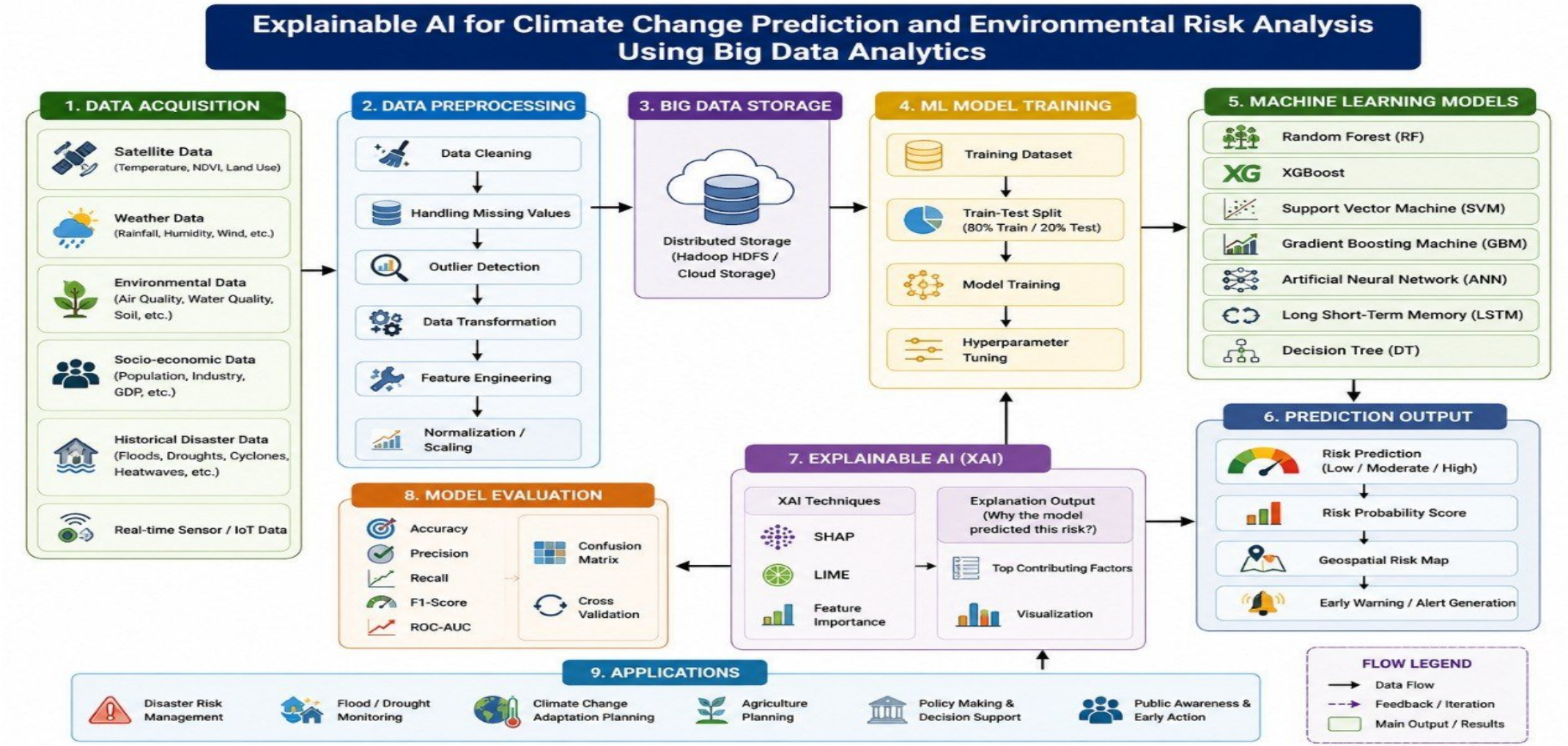
Innovation/ Novelty:

- **Historical + Real-Time Risk Assessment** – Combines historical environmental patterns with live weather and air-quality data.
- **Multi-Factor Risk Analysis** – Uses temperature, humidity, rainfall, wind, pressure, solar radiation, and seasonal features.
- **Data-Driven Risk Scoring** – Converts historical disaster patterns into an interpretable environmental risk score.
- **Explainable AI (XAI)** – Shows how individual environmental features contribute to the prediction.
- **Integrated Streamlit Dashboard** – Connects API data, ML prediction, risk scoring, and explanations in one platform.

"Integration of Three Heterogeneous Datasets for Unified Environmental Disaster Risk Prediction."



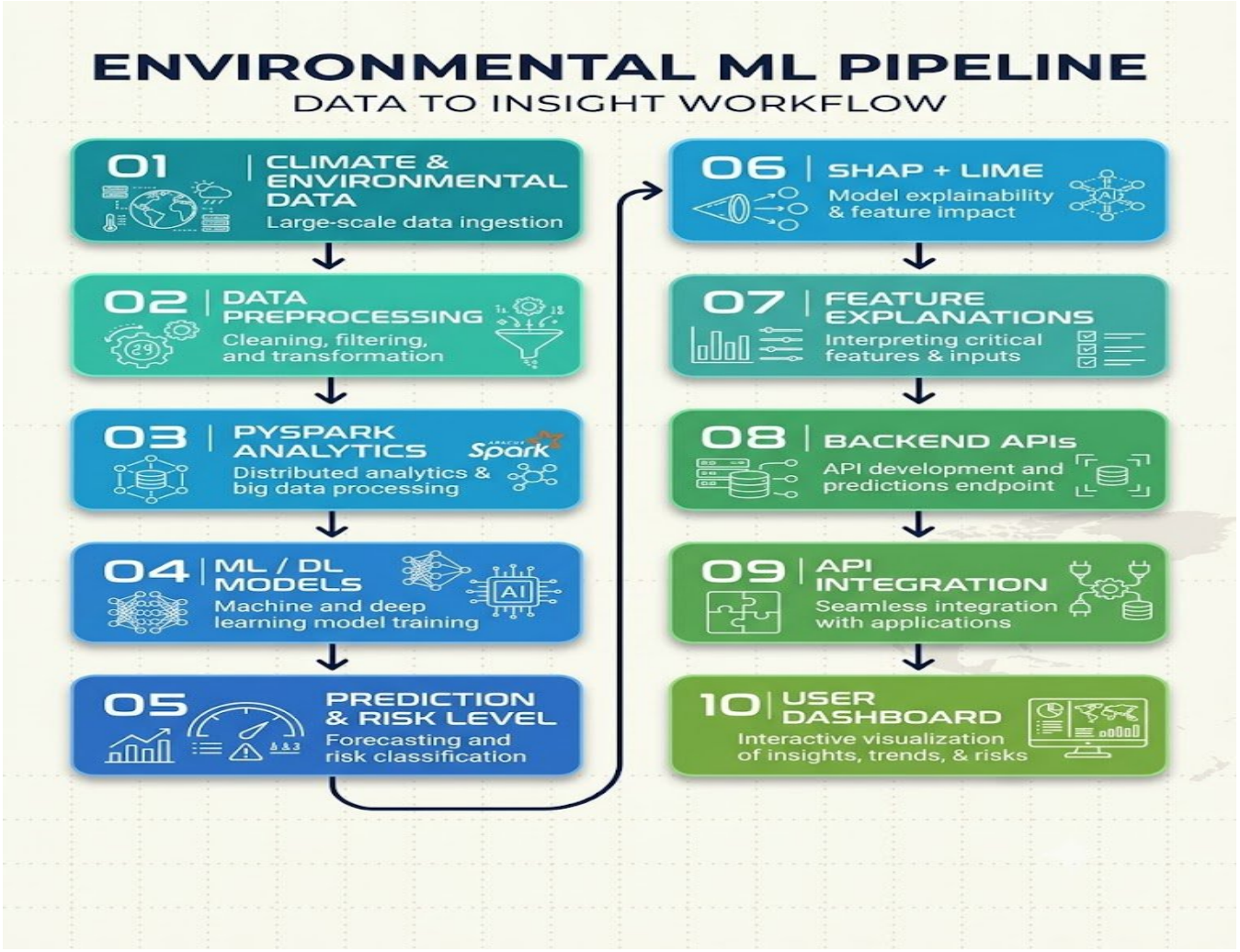
System Design and System Architecture



System Layers:

- Data Layer
- Processing & Analytics Layer
- ML/DL Prediction Layer
- XAI Layer
- API Integration Layer
- Visualization Layer

The framework combines Big Data processing, prediction, risk classification, XAI, and a dashboard.



Project Difficulties & Error Handling:

- **Dataset Integration** → Different datasets required cleaning and feature alignment.
- **Class Imbalance** → Applied imputation and class-balancing techniques.
- **Feature Mismatch** → Ensured identical training and prediction features.
- **Kurnool Prediction Issue** → High ML probability led to investigation and addition of a historical risk-scoring layer.
- **API & Unit Issues** → Handled real-time data formatting and hPa–kPa conversion.

Chalenges and How We Handled Them

Challenge	How We Handled It
Dataset Integration	Cleaned the three datasets individually, standardized column formats, aligned common features, and merged them into a single unified dataset for model training.
Class Imbalance	Used median imputation for missing values and applied class balancing using `class_weight='balanced'` in Logistic Regression so the minority disaster class received greater importance during training.
Feature Mismatch	Defined a fixed set of 8 model features and ensured the same feature names, order, preprocessing, and scaling were used during both training and prediction.
Kurnool Prediction Issue	Investigated the unexpectedly high ML probability by comparing the live conditions with historical environmental patterns. Added a historical risk-scoring layer to provide an additional context-based risk assessment.
API & Unit Issues	Integrated the Open-Meteo APIs, mapped live weather fields to the model features, handled missing/invalid API values, and converted pressure from hPa to kPa (hPa ÷ 10) to match the training data.

Implementation:

Data Preparation

Collect and clean environmental datasets and handled missing/inconsistent data

Big Data Processing

Process datasets using PySpark

Perform EDA and feature analysis

Model Development

Train ML/DL prediction models (Random Forest, XGBoost,
Logistic Regression, SVM)

Hybridised Random Forest and Logistic Regression

Generate environmental risk predictions

Explainability

Applied SHAP

Identify influential environmental factors

API Integration

Expose prediction and risk outputs through APIs

Connect ML outputs with the application interface

Dashboard

Display predictions, trends, risk levels, and explanations

Modern Tools & Technologies:

Python — Core programming and data analysis
Pandas & NumPy — Data preprocessing and analysis
PySpark — Large-scale Big Data processing
Scikit-learn — Machine Learning implementation
Random Forest & XGBoost — Predictive modelling
LSTM / DL Models — Time-series prediction
SHAP & LIME — Explainable AI
REST APIs — Communication between ML system and application
API Integration — Connects predictions with the user interface
Power BI / Python / Streamlit — Visualization and dashboard

Candidate Machine Learning Models

The project considered:

- **Model 1**

Logistic Regression

- **Model 2**

Random Forest

- **Model 3**

HistGradientBoosting

Model 4

XGBoost

- **The final ML architecture uses:**

Logistic Regression + XGBoost

Historical Environmental Conditions

Condition	Threshold
Temperature	≥ 28°C
Humidity	≥ 80%
Wind	≥ 3 m/s
Rainfall	≥ 10 mm
Pressure	≤ 96 kPa

Why Logistic Regression?

Logistic Regression provides:

- interpretability,
- probability output,
- relatively simple decision boundary,
- useful linear baseline,
- easier interpretation of feature effects.

Why XGBoost?

XGBoost provides:

- nonlinear learning,
- tree-based interactions,
- ability to capture nonlinear environmental relationships,
- strong tabular-data performance.

Final Hybrid Model:

The final ML model is a probability-level ensemble.
The equation is:
\[P_{hybrid} = 0.85P_{LR} + 0.15P_{XGB} \]
Therefore:

Logistic Regression = 85%
XGBoost = 15%

- **The weights are frozen in the final dashboard artifact.**
- **This is a genuine mathematical hybrid rather than simply running two independent models.**
- **The final paper defines this architecture explicitly.**

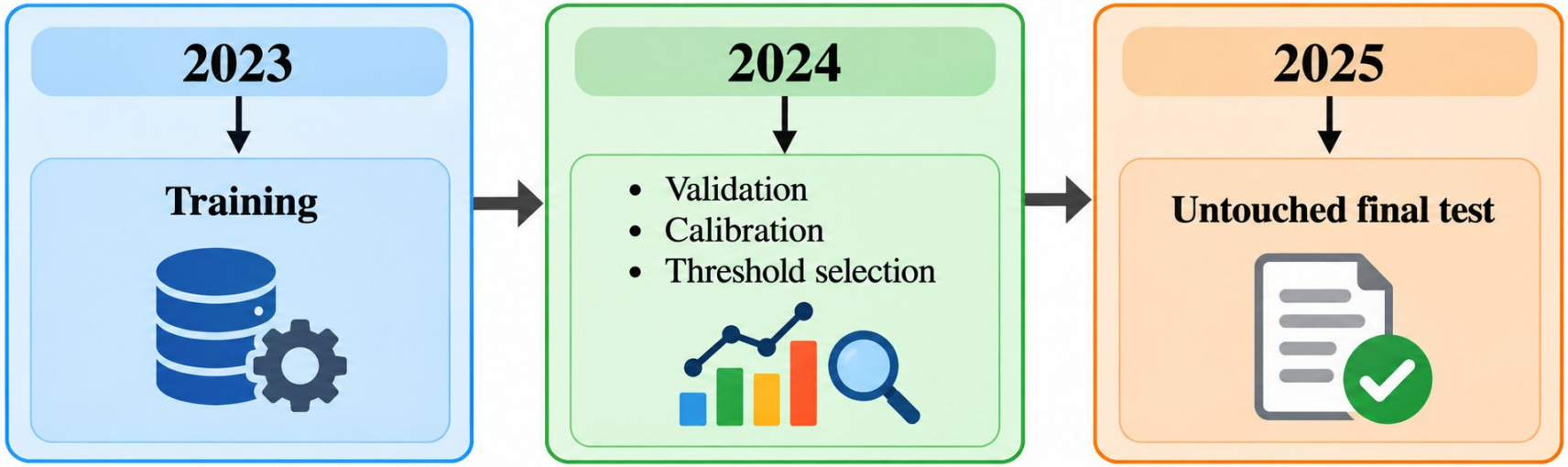
Temporal Evaluation

Instead of using a random split for the final experiment, the project uses a temporal protocol:

2023
↓
Training

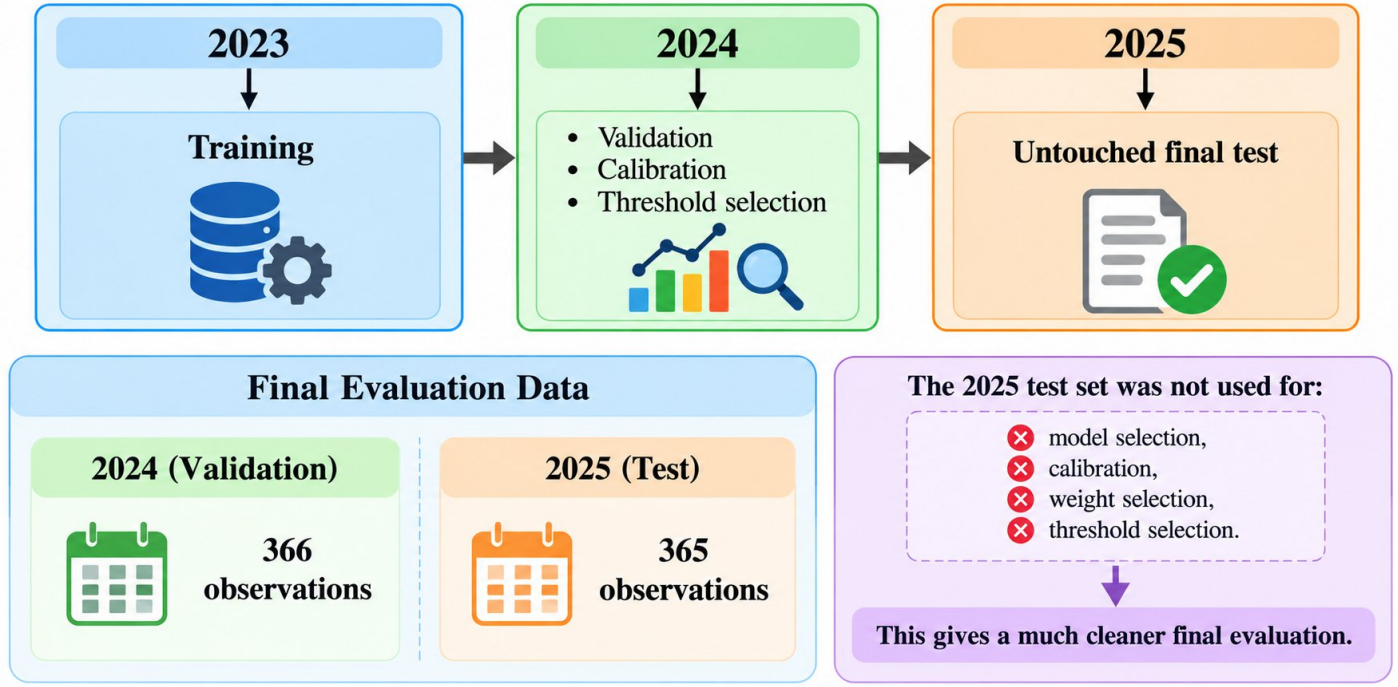
2024
↓
Validation
Calibration
Threshold selection

2025
↓
Untouched final test



The final evaluation contains:
2024
366 observations
2025
365 observations
The 2025 test set was not used for:
- model selection,
- calibration,
- weight selection,
- threshold selection.

This gives a much cleaner final evaluation.



Probability Calibration

The raw hybrid probability is not automatically a reliable probability.

Therefore, the project uses:

Platt Scaling

- Platt scaling maps model scores/probabilities through a sigmoid calibration model.

The calibration is fitted on:

- 2024 validation data and then frozen.
- It is never refitted on the 2025 test set.

ML Decision Threshold

The final calibrated ML threshold is:

$$\boxed{0.24}$$

This threshold was selected using validation data.

It is not:

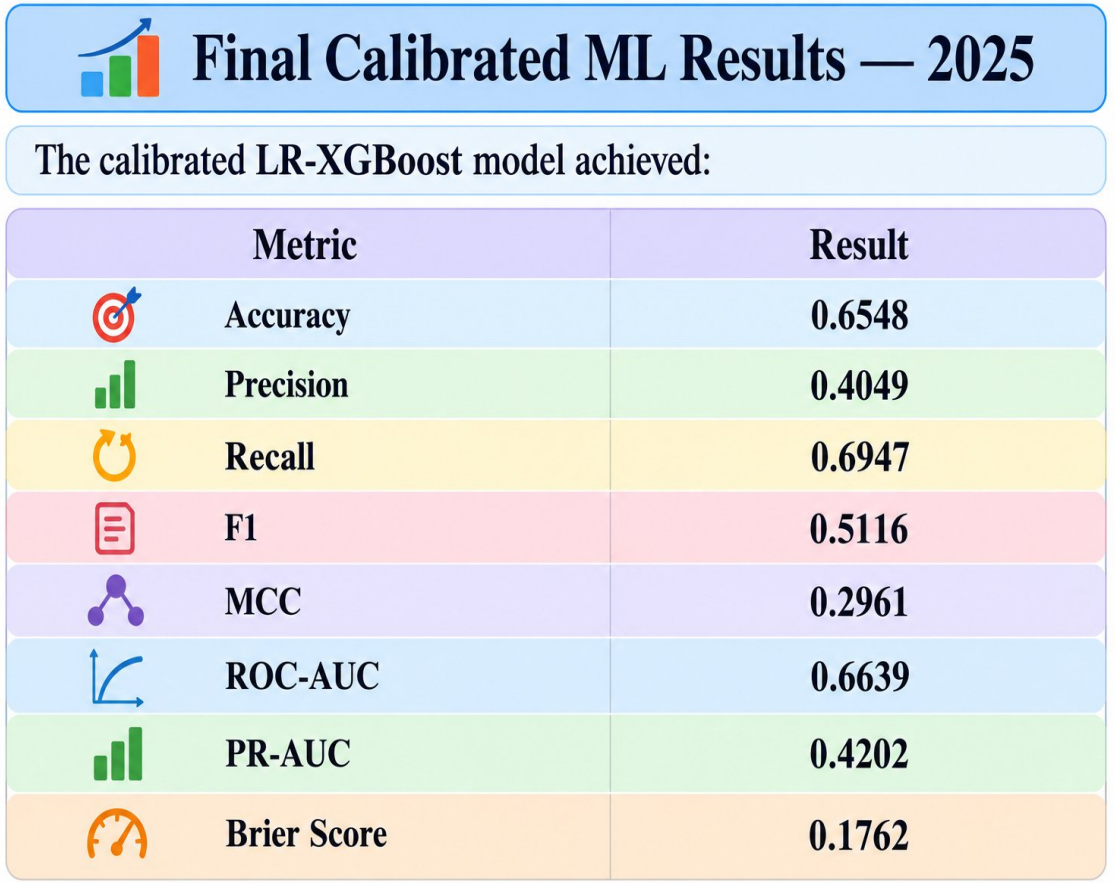
- a physical environmental threshold,
- a temperature threshold,
- a disaster probability established by nature.

It is simply the selected model operating point.

Final Calibrated ML Results — 2025

The calibrated LR-XGBoost model achieved:

Metric	Result
Accuracy	0.6548
Precision	0.4049
Recall	0.6947
F1	0.5116
MCC	0.2961
ROC-AUC	0.6639
PR-AUC	0.4202
Brier Score	0.1762



Confusion matrix:

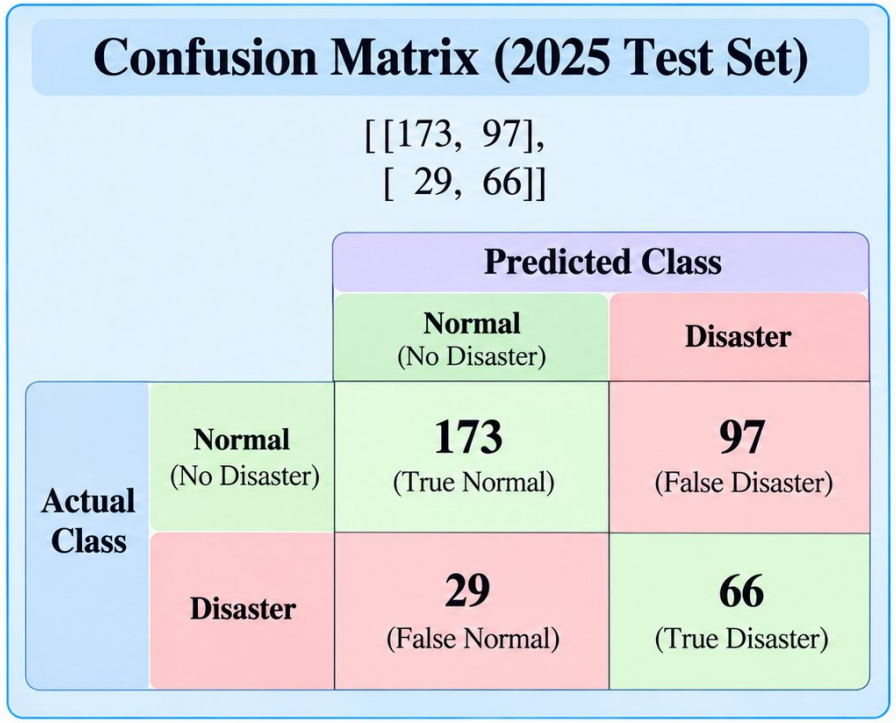
[[173, 97],
[29, 66]]

Meaning:

- True Normal = 173
- False Disaster = 97
- False Normal = 29
- True Disaster = 66

The model detected:

66 of 95 Disaster observations.



Interpretation

- **True Normal (TN) = 173**
  (Actual Normal, Predicted Normal)
- **False Disaster (FP) = 97**
  (Actual Normal, Predicted Disaster)
- **False Normal (FN) = 29**
  (Actual Disaster, Predicted Normal)
- **True Disaster (TP) = 66**
  (Actual Disaster, Predicted Disaster)

Model Detection of Disaster Observations

The model detected
66 of 95
Disaster observations.

Calibration Improvement

Before Platt calibration:

Brier = 0.2246

After Platt calibration:

Brier = 0.1762

Relative reduction:

21.55%

ROC-AUC remained:

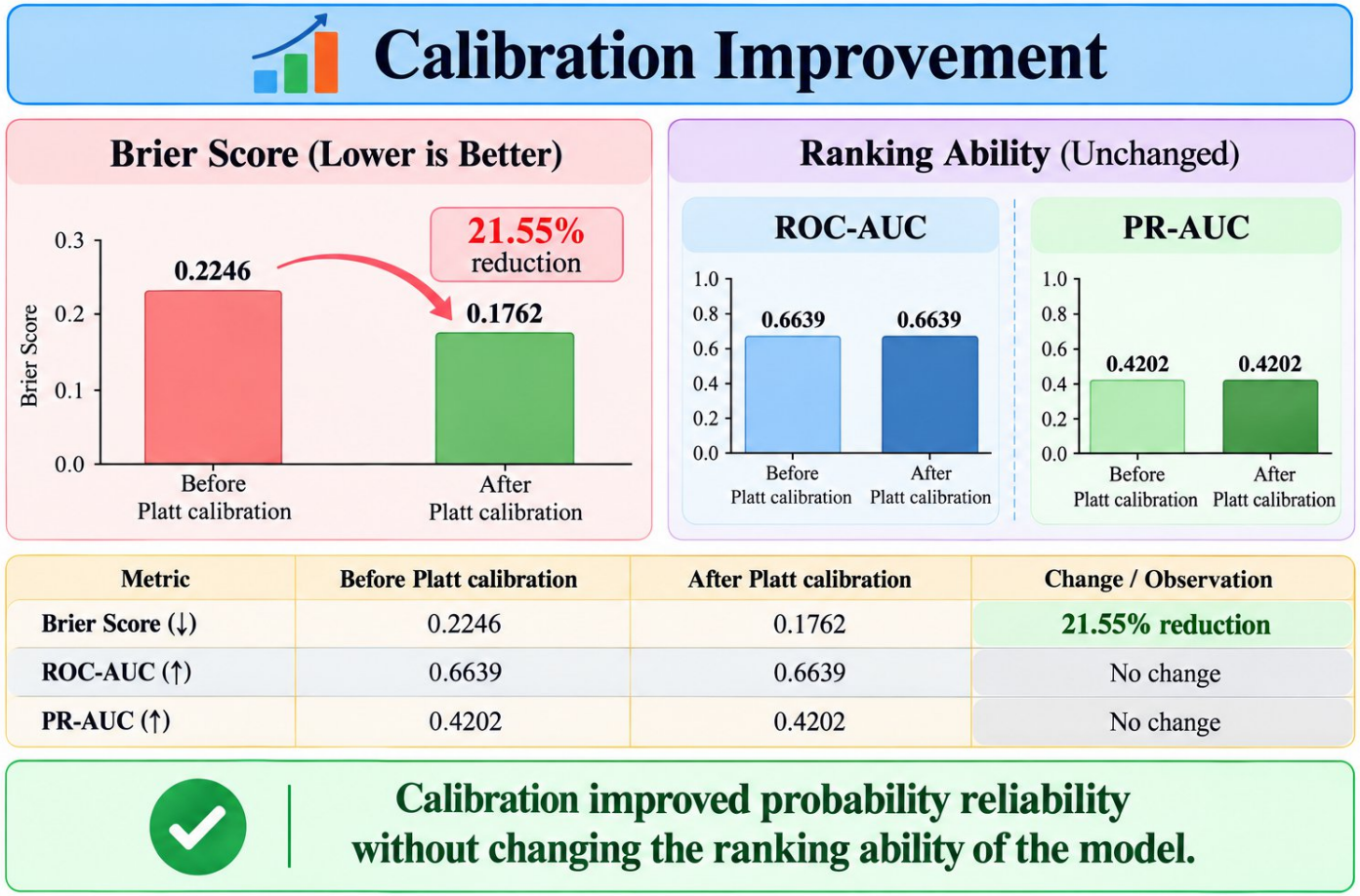
0.6639

PR-AUC remained:

0.4202

Therefore:

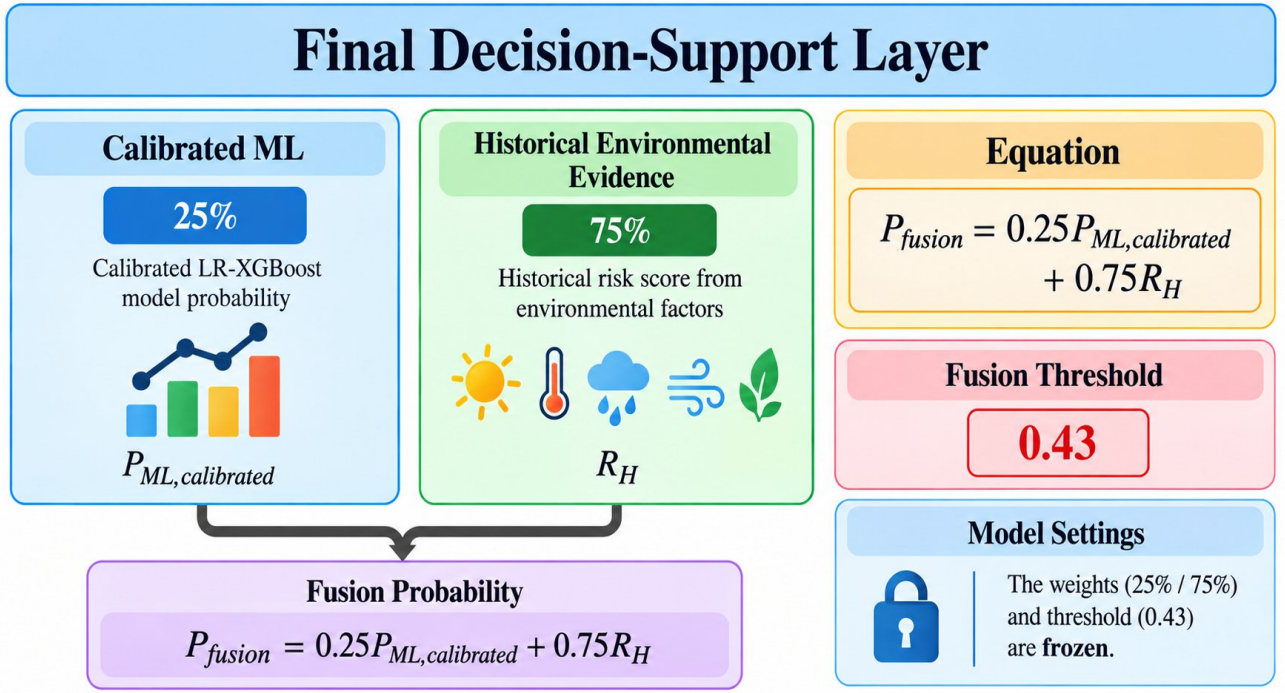
Calibration improved probability reliability without changing the ranking ability of the model.



Metric	Before Platt calibration	After Platt calibration	Change / Observation
Brier Score (↓)	0.2246	0.1762	21.55% reduction
ROC-AUC (↑)	0.6639	0.6639	No change
PR-AUC (↑)	0.4202	0.4202	No change

Calibration improved probability reliability without changing the ranking ability of the model.

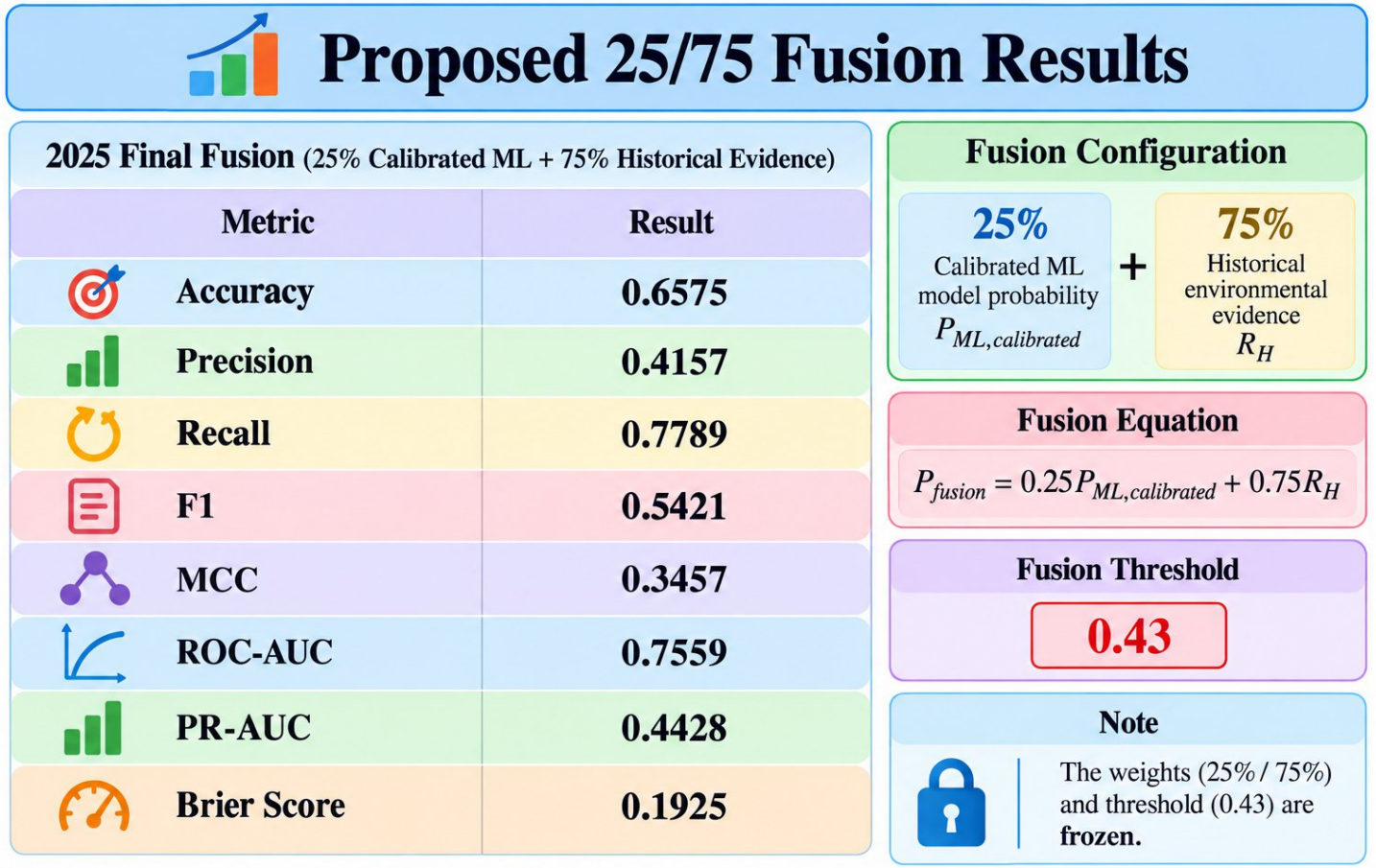
Historical Risk + ML Fusion
The final decision-support layer combines:
25% calibrated ML
and
75% historical environmental evidence.
Equation:
P_{fusion} = 0.25P_{ML,calibrated} + 0.75R_H
Fusion threshold:
boxed{0.43}
The weights and threshold are frozen.



Proposed 25/75 Fusion Results

2025 final fusion:

Metric	Result
Accuracy	0.6575
Precision	0.4157
Recall	0.7789
F1	0.5421
MCC	0.3457
ROC-AUC	0.7559
PR-AUC	0.4428
Brier	0.1925



Compared with the calibrated ML component:
Recall:
0.6947 → 0.7789

F1:
0.5116 → 0.5421

Accuracy:
0.6548 → 0.6575
So the strongest benefit of the fusion is:
better minority-class retrieval, especially recall, while keeping accuracy almost unchanged.

OOD Detection

One of the important innovations is:

Out-of-Distribution Detection

OOD means:

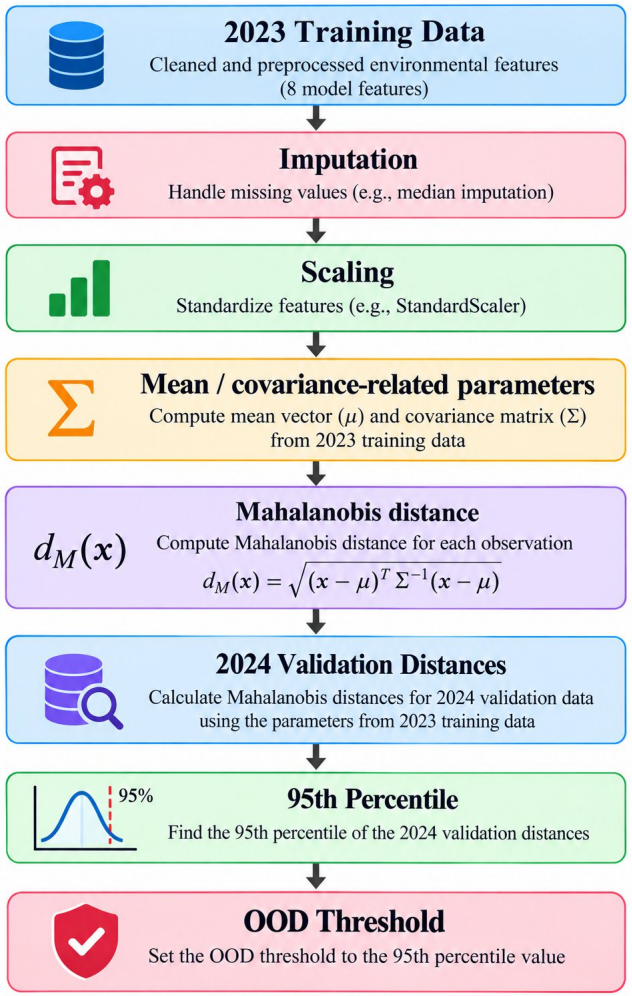
The current observation does not look sufficiently similar to the environmental distribution used during training.

The project uses:

Mahalanobis distance

The OOD detector is fitted using the training distribution.

2023 Training Data
↓
Imputation
↓
Scaling
↓
Mean / covariance-related parameters
↓
Mahalanobis distance
↓
2024 validation distances
↓
95th percentile
↓
OOD threshold



Results:

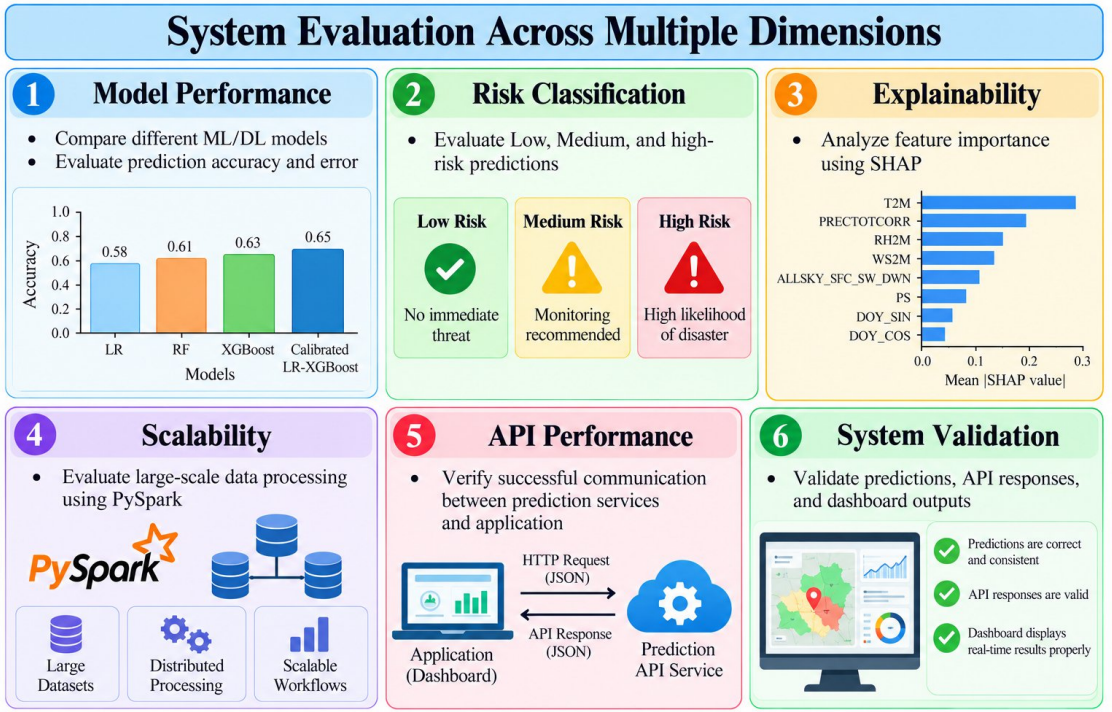
- Processed and analyzed climate/environmental data
- Identified patterns and relationships among environmental variables
- Generated climate and environmental risk predictions
- Classified environmental conditions into:
  - **Low Risk**
  - **Medium Risk**
  - **High Risk**
- Identified important factors influencing predictions
- Generated interpretable explanations using SHAP and LIME
- Integrated prediction results with the application through APIs
- Presented results through an interactive dashboard

Final Model Metrics (Hybridized Logistic Regression)	
Metric	**Value**
Accuracy	**86%**
Precision	**0.3919**
Recall	**0.5686**
F1-score	**0.464**
ROC-AUC	**0.8133**
Decision Threshold	**0.71**
Test Samples	**485**
Confusion Matrix	**[[389, 45], [22, 29]]**

Performance Evaluation:

The system is evaluated across multiple dimensions:

- **Model Performance**
  - Compare different ML/DL models
  - Evaluate prediction accuracy and error
- **Risk Classification**
  - Evaluate Low, Medium, and high-risk predictions
- **Explainability**
  - Analyze feature importance using SHAP
- **Scalability**
  - Evaluate large-scale data processing using PySpark
- **API Performance**
  - Verify successful communication between prediction services and application
- **System Validation**
  - Validate predictions, API responses, and dashboard outputs

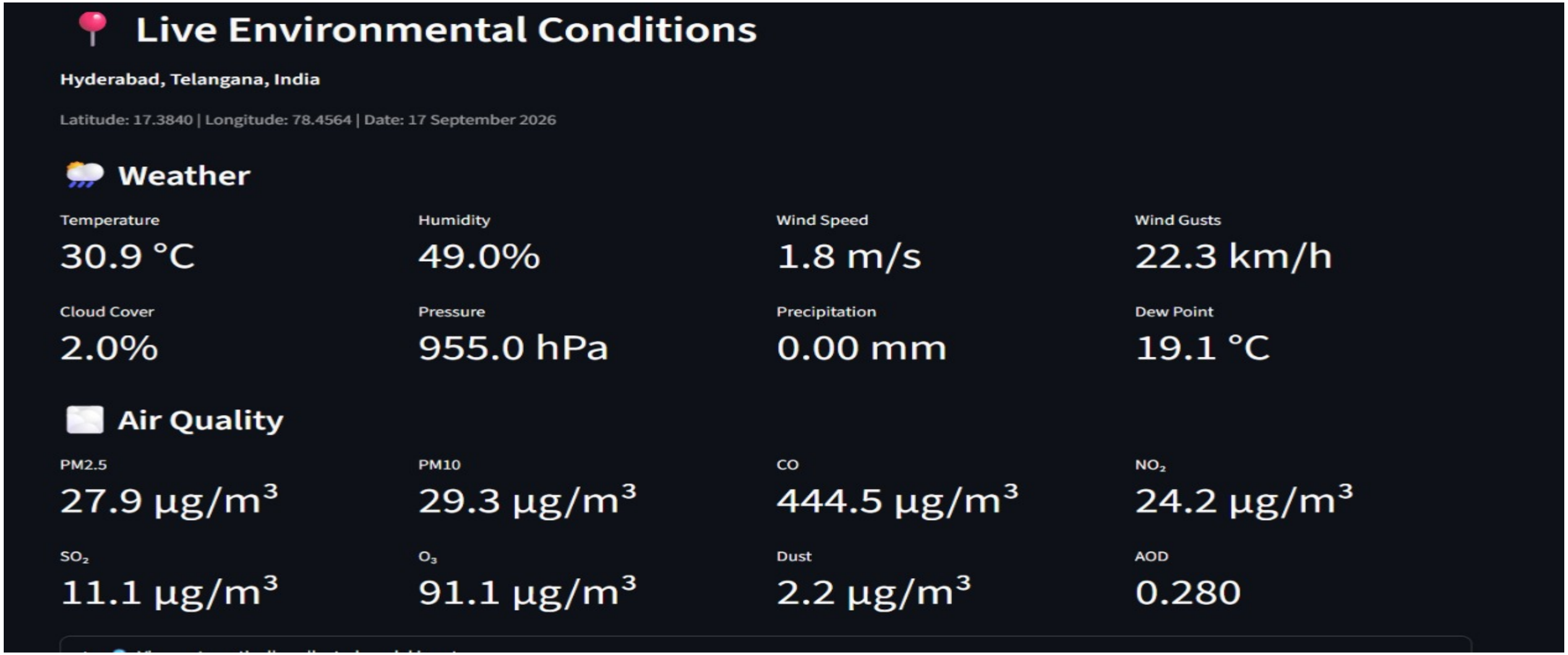


Project Outcome:



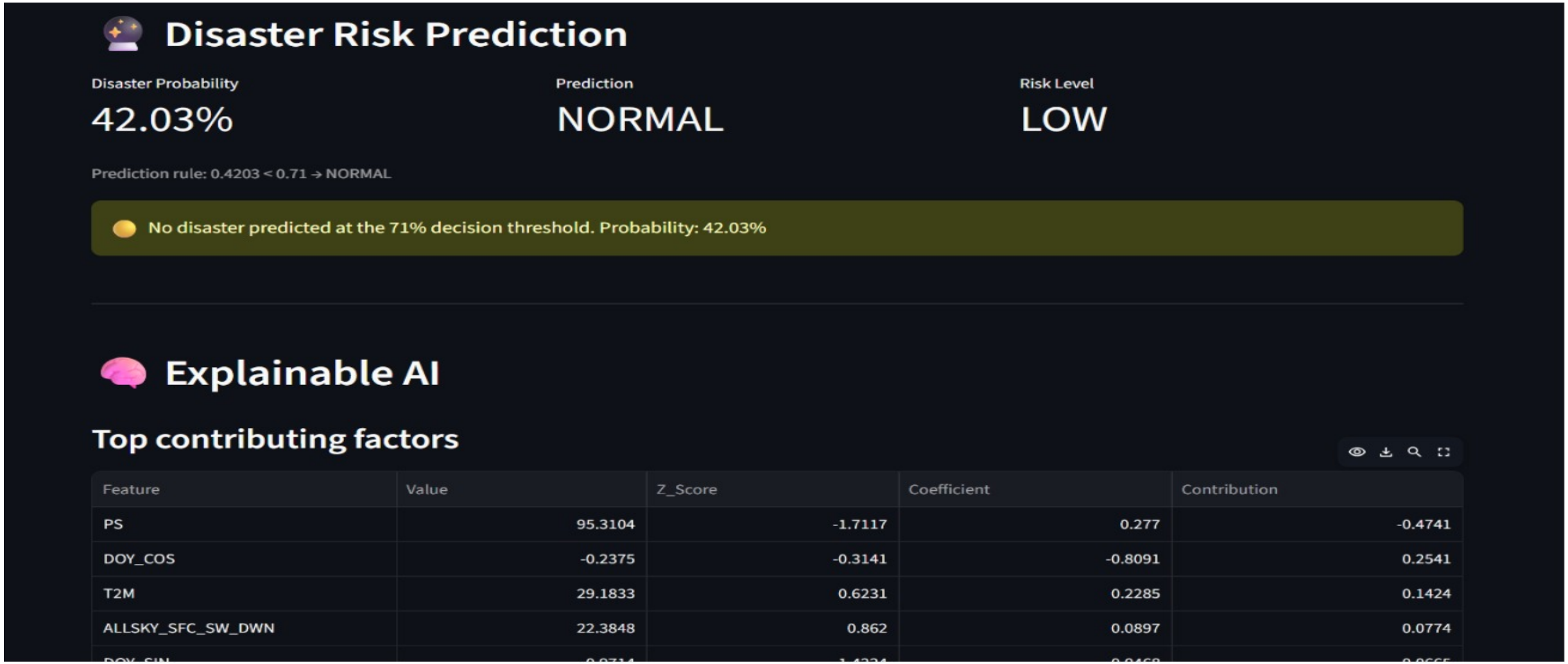
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Project Outcome:



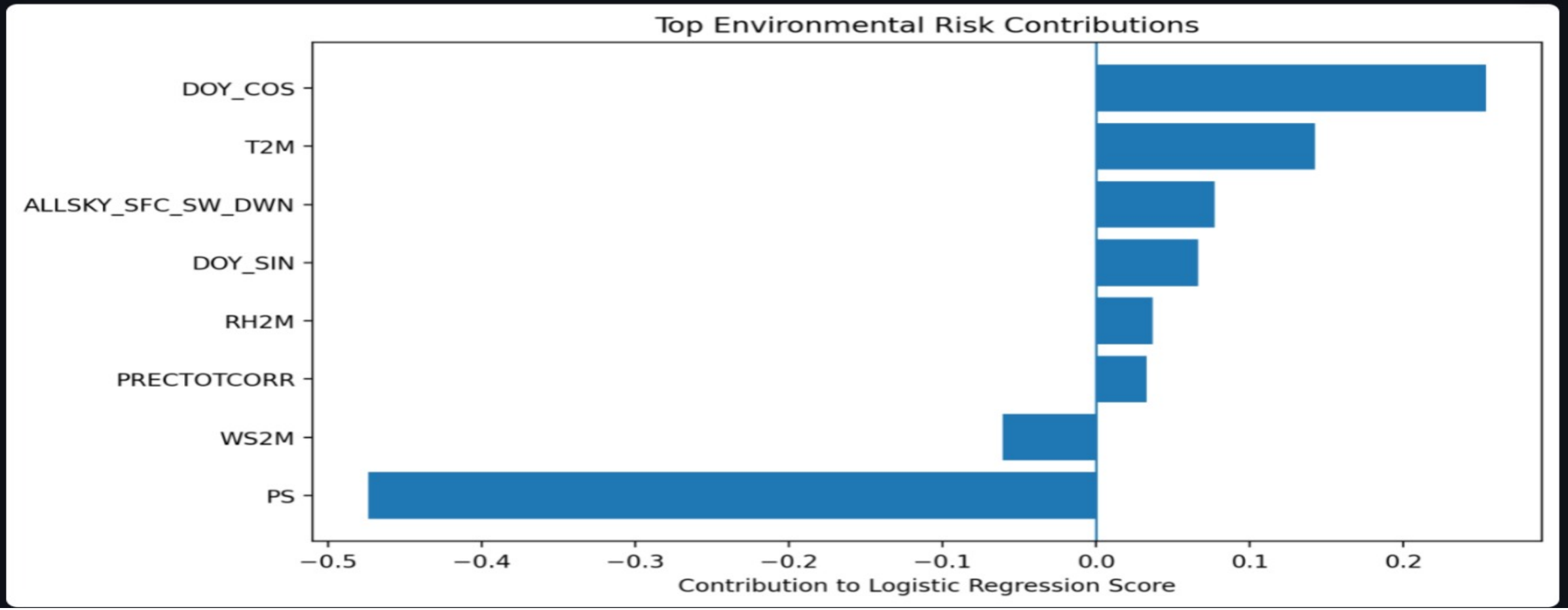
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Project Outcome:

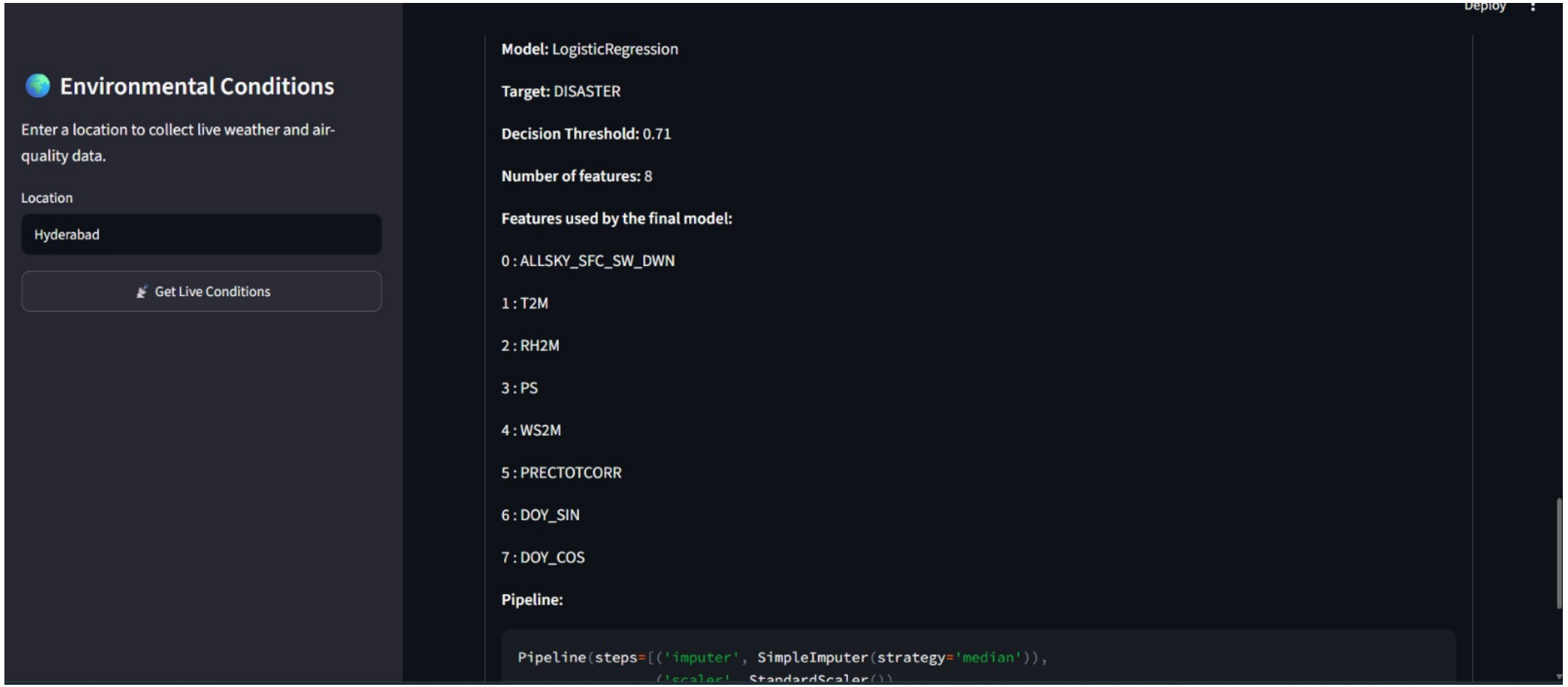


Project Outcome:

Risk contribution chart



Project Outcome:



Validating Methodology:

- Followed a structured pipeline from data integration to final evaluation
- Applied temporal validation to prevent data leakage
- Used 2023 for training, 2024 for validation & calibration, and 2025 as an unseen test set
- Evaluated temporal generalization on future data
- Integrated ML prediction, historical risk, calibration, OOD detection, and explainability



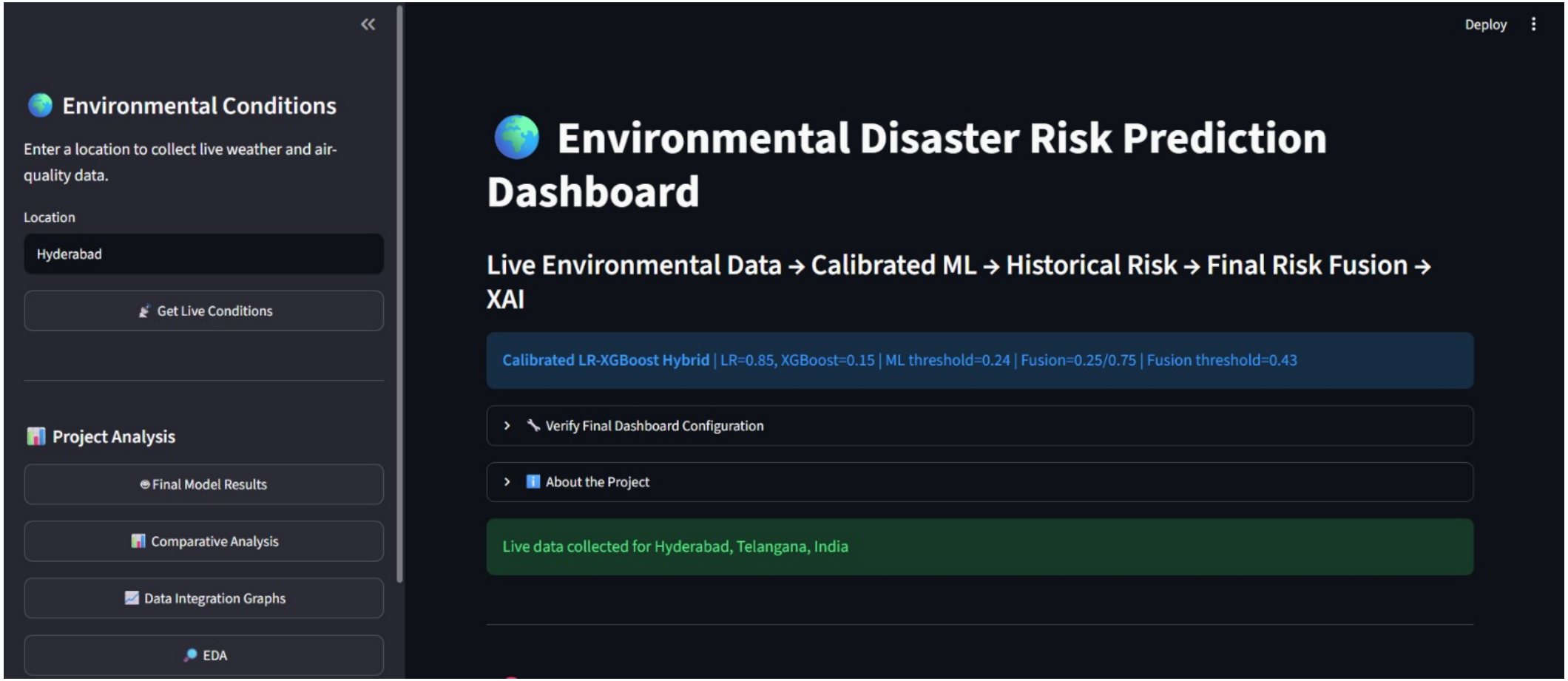
Code Implementation:

- Developed completely in Python
- Major libraries: Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Matplotlib, Streamlit
- Implemented:
  - Data integration & preprocessing
  - Feature engineering & model training
  - Probability calibration
  - Historical-risk calculation & risk fusion
  - OOD detection using Mahalanobis distance
  - SHAP-based explainability
- Verified model artifacts, feature ordering, and prediction pipeline

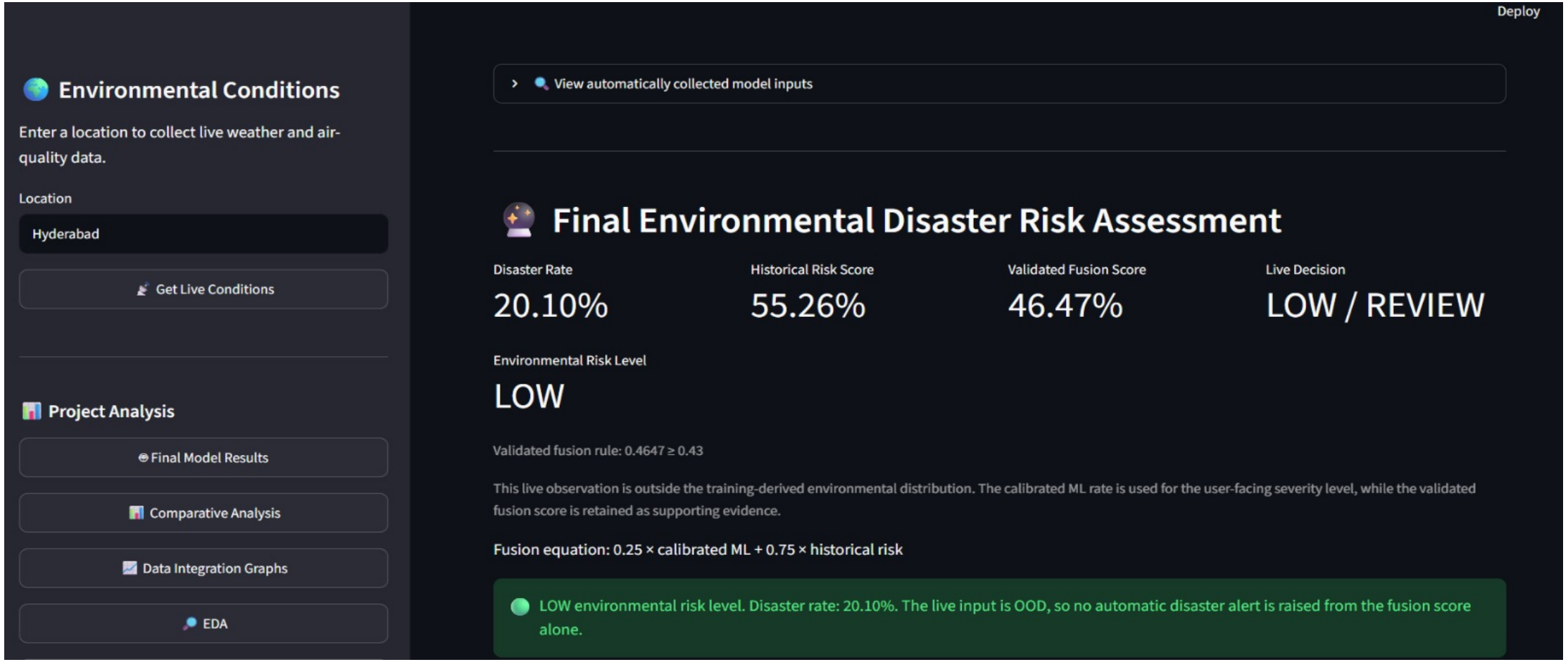
Dashboard Execution:

- Environmental Disaster Risk Prediction Dashboard
- Developed using Streamlit
- Accepts location-based input
- Retrieves live weather and air-quality data
- Generates calibrated ML risk prediction
- Calculates historical environmental risk
- Performs ML + historical risk fusion
- Checks for distribution shift
- Displays model results, EDA, comparisons & XAI

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Final Model and Ablation Results

The final evaluation uses a temporal protocol: 2023 for training, 2024 for validation/calibration, and 2025 as the untouched final test set.

Configuration	Accuracy	Precision	Recall	F1	MCC	ROC-AUC	PR-AUC	Brier
Calibrated LR-XGBoost	0.6548	0.4049	0.6947	0.5116	0.2961	0.6639	0.4202	0.1762
Historical Risk	0.6575	0.4148	0.7684	0.5387	0.3398	0.7574	0.4403	0.2080
Proposed 25/75 Fusion	0.6575	0.4157	0.7789	0.5421	0.3457	0.7559	0.4428	0.1925

Final calibrated ML configuration

Parameter	Value
Logistic Regression weight	0.85
XGBoost weight	0.15
ML calibration	Platt / sigmoid scaling
ML threshold	0.24
Fusion ML weight	0.25
Fusion historical weight	0.75
Fusion threshold	0.43
OOD threshold	16.0946

Calibration result

Platt scaling reduced the 2025 Brier score from **0.2246** to **0.1762**, a **21.55% relative reduction** in probabilistic error. ROC-AUC remained **0.6639** and PR-AUC remained **0.4202**, as expected for a monotonic probability calibration step.

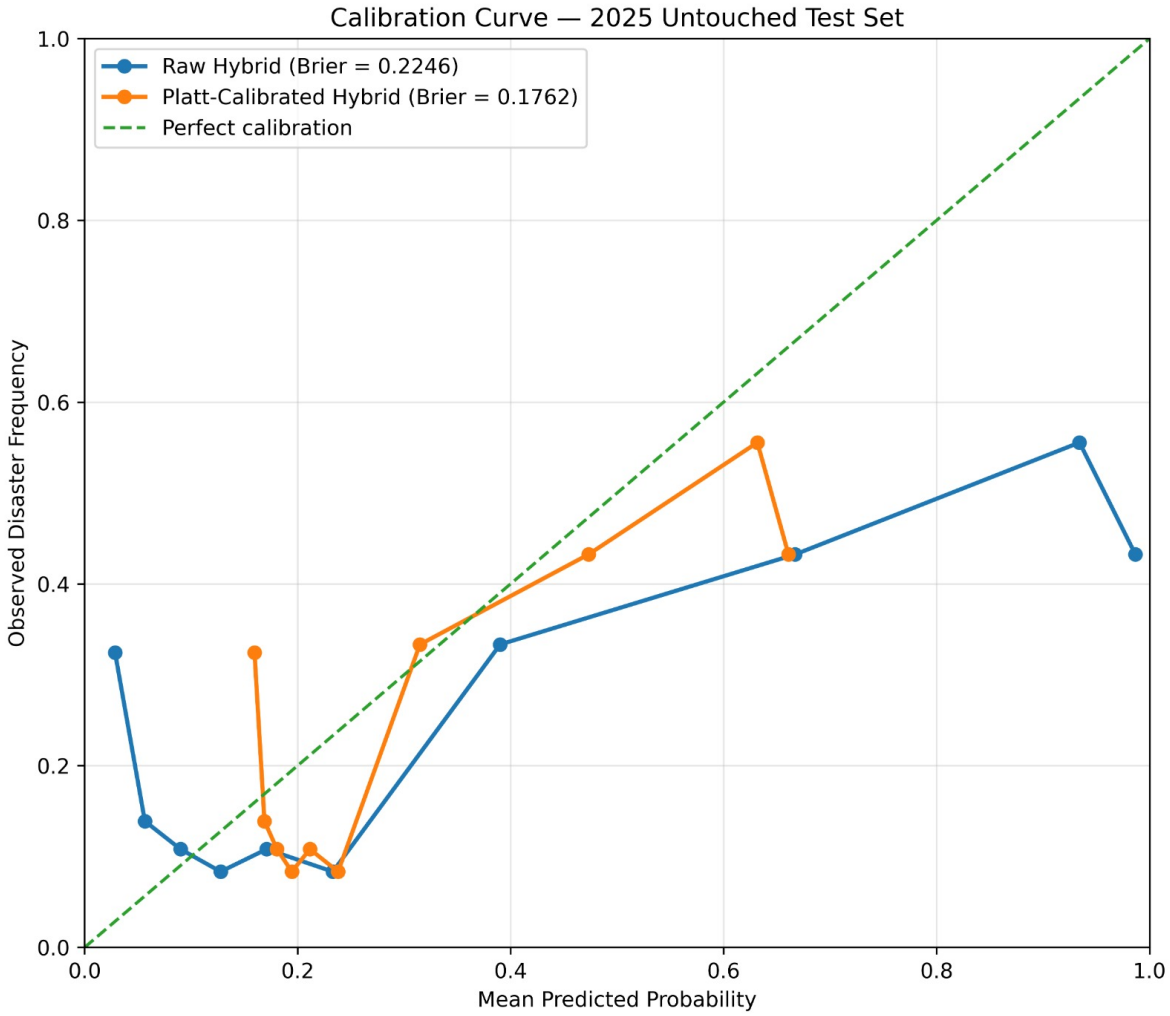
OOD reliability result

Subset	Samples	Brier	Accuracy	Recall	MCC
2025 In-distribution	344	0.1720	0.6657	0.6588	0.2863
2025 OOD	21	0.2444	0.4762	1.0000	0.0000

Comparative Analysis of Baseline Models:

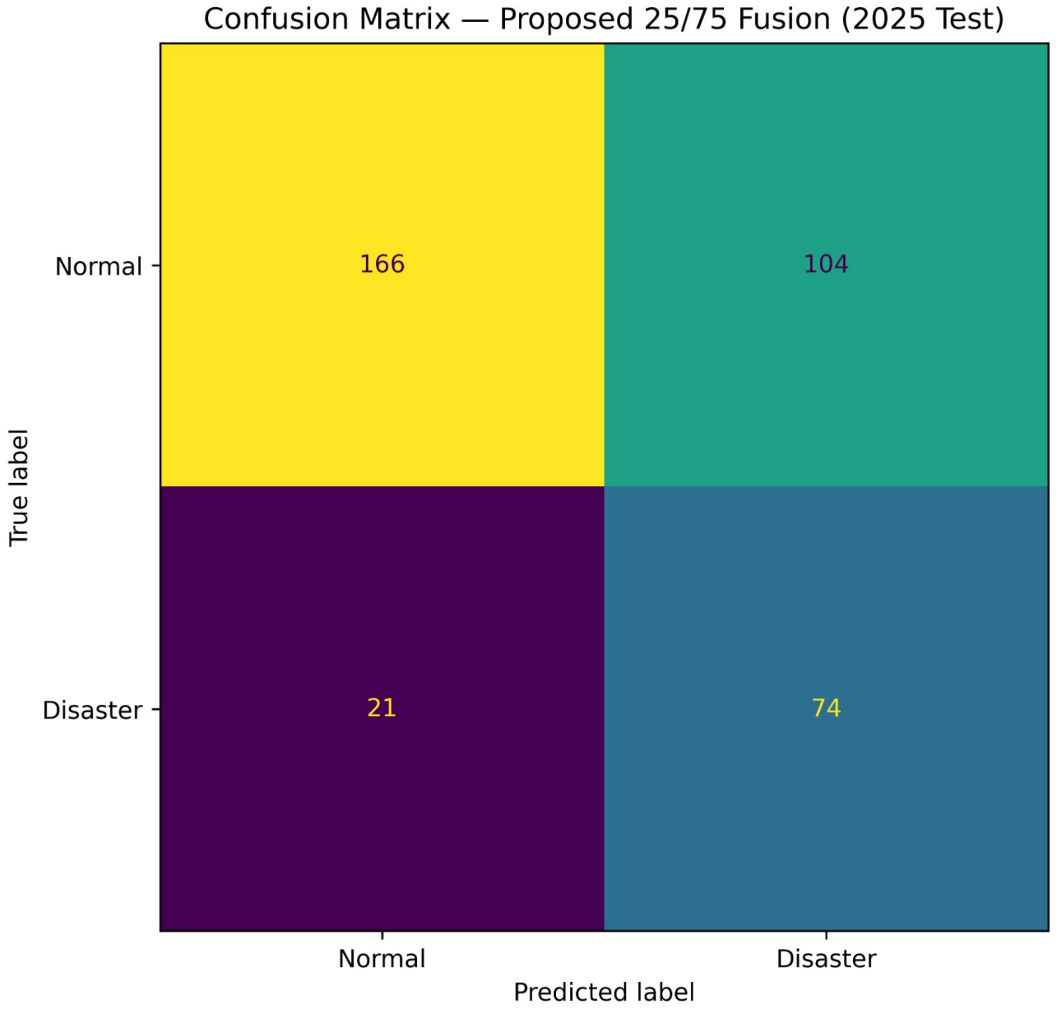
- A comparative analysis evaluates the performance of different machine-learning approaches
- The dashboard compares four models:
  - Logistic Regression
  - Random Forest
  - XGBoost
  - Proposed Hybridized Logistic Regression + XGBoost
- Metrics used: Accuracy, Precision, Recall, F1-score, ROC-AUC
- The proposed hybrid model is visually highlighted in the dashboard to set it apart from the baselines and show it is the final proposed approach

Results – Calibration Curve (2025 Untouched Test Set):



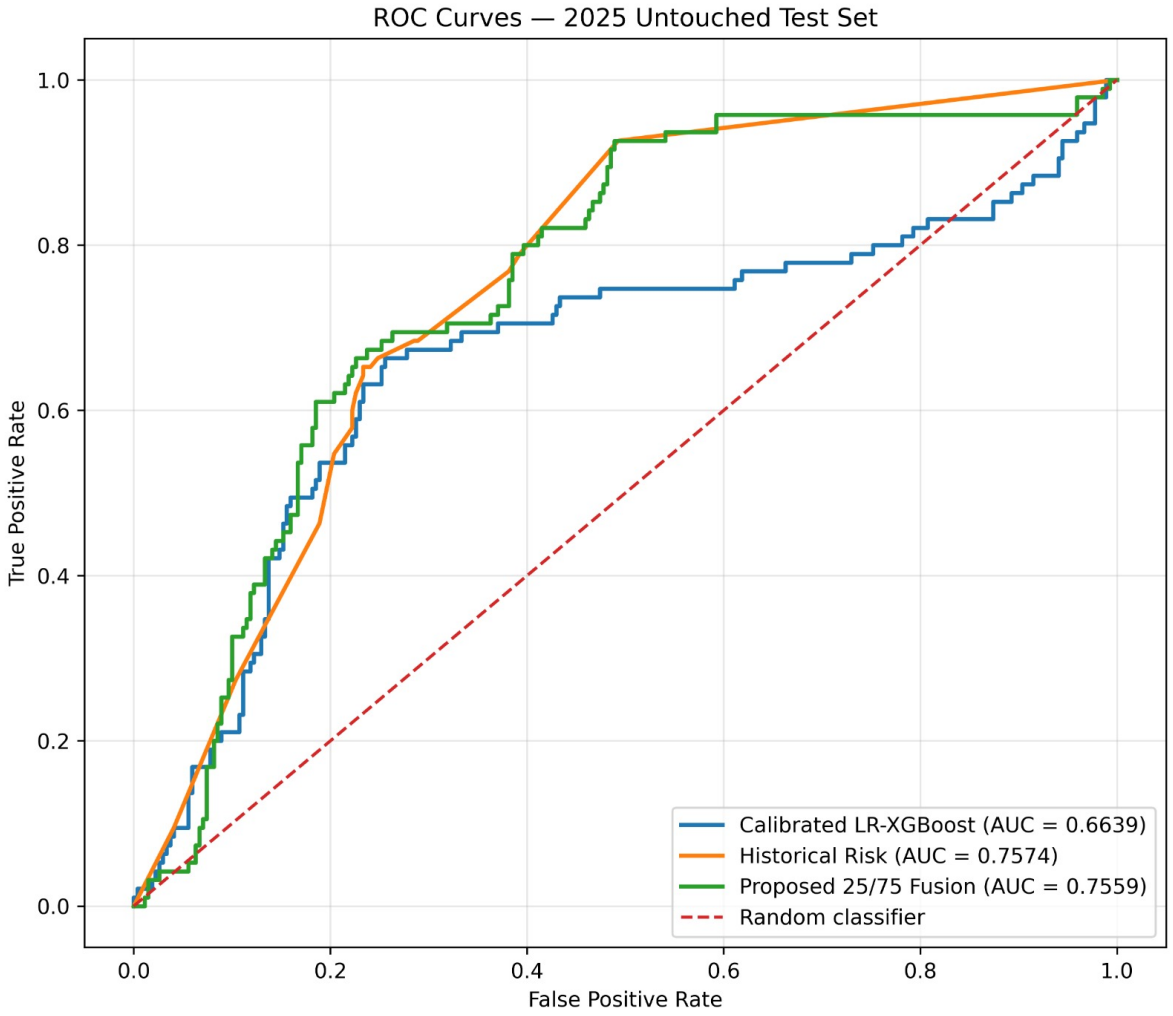
- Platt scaling lowers the Brier score from 0.2246 (raw hybrid) to 0.1762, so predicted probabilities are more reliable.
- The raw hybrid is overconfident at high scores: predictions near 0.93–0.99 correspond to only about 0.43–0.55 observed disaster frequency.
- The calibrated curve sits closer to the perfect-calibration diagonal across the mid-to-high probability range.

Results – Confusion Matrix, Proposed 25/75 Fusion (2025 Test):



- TN = 166, FP = 104, FN = 21, TP = 74 (365 test samples).
- Recall = 77.9% (74 of 95 disaster cases detected); Precision = 41.6%.
- Accuracy = 65.8%, F1-score = 0.542, Specificity = 61.5%.
- Only 21 disasters are missed, at the cost of more false alarms, which suits an early-warning use case.

Results – ROC Curves (2025 Untouched Test Set):



- Calibrated LR-XGBoost alone: AUC = 0.6639.
- Historical risk layer alone: AUC = 0.7574.
- Proposed 25/75 fusion: AUC = 0.7559, a gain of about 0.092 over the ML model alone and on par with the historical layer.
- Fusion keeps the learned ML signal while adding the interpretable historical layer.

GitHub Repository:

- Updated project repository with:
- Source code
- Streamlit dashboard
- Dataset-processing components
- Model artifacts
- Documentation
- Project outputs

GitHub Link: https://github.com/mrudani1712/KLH-CSE-2026-27-2420030591-Climate-Risk-XAI.git

Research Paper Draft Completion:

- Draft prepared based on the completed implementation and experiments
- Includes:
  - Three-dataset integration
  - Preprocessing & feature engineering
  - Temporal evaluation
  - Hybrid ML model
  - Probability calibration
  - Historical risk layer
  - 25/75 risk fusion
  - OOD analysis
  - SHAP explainability
  - Dashboard implementation
- Figures, results and literature references incorporated

Thank You